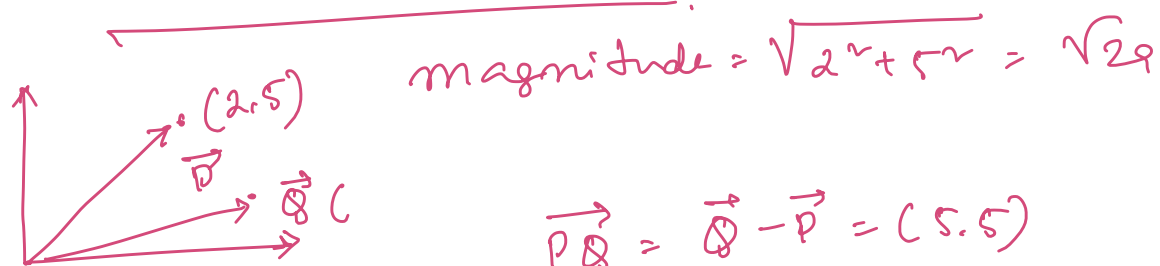


Support Vector Machine

Coordinate Geometry

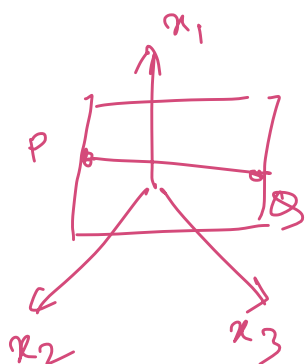


$$\vec{PQ} = \vec{Q} - \vec{P} = (5, 5)$$

$$\vec{Q} (7, 10)$$

$$\vec{P} (2, 5)$$

$$\text{Angle } \tan \theta = y/x \Rightarrow \theta = \tan^{-1}(y/x)$$



$$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0$$

$$w^T x + w_0 = 0$$

$$w = \{w_1, w_2, \dots, w_3\}$$

$$x = \{x_1, x_2, x_3\}$$

Normal vector of $f(x)$

$$\vec{w} = [w_1, w_2, w_3] \Rightarrow \nabla f(x)$$

show that $\vec{PQ} \cdot \vec{w} = 0$

$$\begin{aligned} \vec{P} &= w_1 p_1 + w_2 p_2 + w_3 p_3 + w_0 \Rightarrow w^T P = -w_0 \\ \vec{Q} &= w_1 q_1 + w_2 q_2 + w_3 q_3 + w_0 \Rightarrow w^T Q = -w_0 \end{aligned}$$

$$\begin{aligned} w^T P - w^T Q &= 0 \\ \Rightarrow w^T (P - Q) &= 0 \\ &= \underline{w^T \cdot \vec{PQ} = 0} \end{aligned}$$

distance of a point x from the plane. $w^T x + w_0 = 0$

$$d = \frac{w^T x + w_0}{|w_0|} = \gamma = \underline{\text{geometric distance}}$$

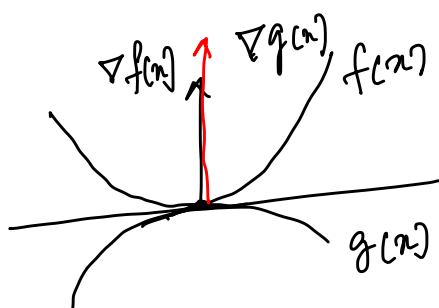
Functional distance $w^T x + w_0 = \hat{\gamma}$

04/09/25

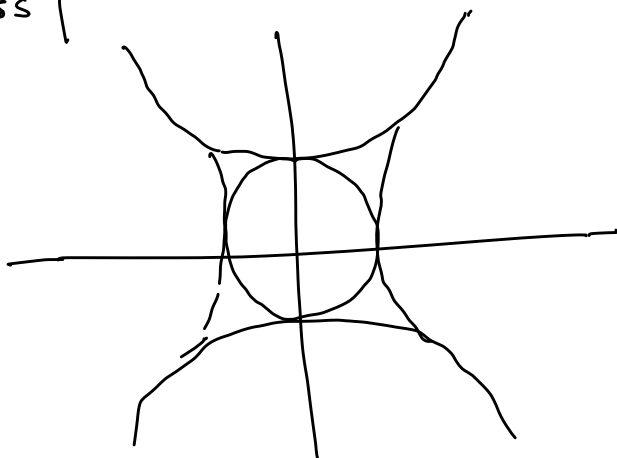
Lagrange Multiplier

$$\begin{aligned} \max_{x, y} \quad & x^v y^v \Leftarrow \\ \text{s.t.} \quad & \underline{x^v + y^v = 1} \end{aligned}$$

Regression - linear - Sq loss
- logistic - log loss



$$\boxed{\nabla f(x) = \lambda \nabla g(x)}$$



$\Rightarrow f(x) = x^2 + y^2 - 4$
 at $(2,0)$
 $\nabla f(x) = (2x, 2y)$
 at $(2,0)$ $(4,0)$

$g(x) = \frac{x^2}{4} + y^2 - 1$: Ellipse
 $\nabla g(x) = (x/2, 2y)$
 at $(2,0)$ $(4,0)$

Lagrange Ex. $L(x, \lambda) = f(x) - \lambda g(x)$ λ = Lagrange multiplier.

max $f(x)$
 s.t. $g(x)$
equality

and $\nabla_x L = 0$

$\nabla_{\lambda} L = 0$

$f(x) - \lambda_1 g_1(x) - \lambda_2 g_2(x)$

$\nabla_x L = 0$

$\nabla_{\lambda_i} L = 0$

max $f(h, s) = 200h^{2/3}s^{1/3}$

s.t. $c \quad 20h + 170s = 20000$

$L(h, s, \lambda) = 200h^{2/3}s^{1/3} - \lambda(20h + 170s - 20000)$

$\frac{\partial L}{\partial h} = 200 \times \frac{2}{3} h^{-1/3} s^{1/3} - 20\lambda = 0$

$\frac{\partial L}{\partial s} = 200 \times \frac{1}{3} h^{2/3} s^{-2/3} - 170\lambda = 0$

$h = 666.66$

$s = 39.12$

$\lambda = 3.59$

$\frac{\partial L}{\partial \lambda} = -20h - 170s + 20000 = 0$

$f() = 51777$
max

KKT Theory
 Karush Kuhn Tucker

max $f(x)$

s.t. $h_i(x) \leq 0$

max: $-x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2$

s.t. $x_1 + x_2 \leq 2$

$2x_1 + 3x_2 \leq 12$

$x_1, x_2 \geq 0$

① Convert it to LM
 take derivative w.r.t. variable
 and equate them to 0

② $\lambda_i h_i = 0$ *

③ $h_i \leq 0$

④ $\lambda_i \geq 0$

Cond¹: $L() = -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2 - \lambda_1(x_1 + x_2 - 2) - \lambda_2(2x_1 + 3x_2 - 12)$

$\frac{\partial L}{\partial x_1} = -2x_1 + 4 - \lambda_1 - 2\lambda_2 = 0$

①

$$\frac{\partial L}{\partial x_2} = -2x_2 + 6 - \lambda_1 - 3\lambda_2 = 0 \quad \text{--- (1b) ✓}$$

$$\frac{\partial L}{\partial x_3} = -2x_3 = 0 \Rightarrow x_3 = 0$$

Condⁿ

$$\lambda_1 (x_1 + x_2 - 2) = 0 \quad \text{--- (2a) ✓}$$

$$\lambda_2 (2x_1 + 3x_2 - 12) = 0 \quad \text{--- (2b) ✓}$$

Cond³

$$x_1 + x_2 - 2 \leq 0 \quad \text{--- (3a) ✓}$$

$$2x_1 + 3x_2 - 12 \leq 0 \quad \text{--- (3b) ✓}$$

$$\text{Cond 4: } \lambda_1, \lambda_2 \geq 0$$

Case 1: $\lambda_1 = 0, \lambda_2 = 0$ (1a) (1b) $x_1 = 2, x_2 = 3$
 $\times$ (3a) $3 \leq 0 \times$

Case 2: $\lambda_1 \neq 0, \lambda_2 \neq 0$; $x_1 + x_2 - 2 = 0$
 $2x_1 + 3x_2 - 12 = 0 \Rightarrow \begin{cases} x_1 = 8 & x_2 = -6 \\ \lambda_2 = -26 \end{cases}$
 (1a) (1b) $\lambda_2 = -26 \times$

Case 3: $\lambda_1 = 0, \lambda_2 \neq 0$ (1a) (1b) $-2x_1 + 4 - 2\lambda_2 = 0$
 $-2x_2 + 6 - 3\lambda_2 = 0 \Rightarrow \begin{cases} x_1 = \frac{2}{3}x_2 \\ 2x_1 + 3x_2 - 12 = 0 \end{cases}$
 $\Rightarrow x_1 = 2, x_2 = 3$
 (2a) $5 - 2 \leq 0 \times$

Case 4: $\lambda_1 \neq 0, \lambda_2 = 0$ $x_1 = 4, x_2 = 3/2, \lambda_2 = 0, \lambda_1 = 3$

Primal and Dual theorem

min_w $f(w)$ ✓

w.r.t. $g_i(w) \leq 0 \quad i=1 \dots k$ ✓
 $h_i(w) = 0 \quad i=1 \dots l$

Generalized Lagrange Eqⁿ: $L(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w)$

let us define: $\underline{\underline{D_p(w)}} = \max_{\alpha, \beta} L(w, \alpha, \beta)$

$$= \max_{\substack{\alpha_i \geq 0 \\ \beta_i \text{ free}}} f(w) + \sum \alpha_i g_i(w) + \sum \beta_i h_i(w)$$

Condⁿ Violated

$$g_i(w) > 0 \quad \alpha_i \rightarrow \infty \quad D_p(w) \rightarrow \infty$$

$$h_i(w) \neq 0 \quad \beta_i \rightarrow +\infty / -\infty \quad D_p(w) \rightarrow \infty$$

Condⁿ Satisfied

$$D_p(w) = f(w) \quad \left| \quad D_p(w) = \begin{cases} f(w) & \text{Satisfied} \\ \infty & \text{otherwise} \end{cases}$$

Primal

Soln of the
primal
problem

$$p^* = \min_w \max_{\alpha, \beta} L(w, \alpha, \beta)$$

max-min
inequality
thm

$$\max \min f(w) \leq \min \max f(w)$$

$$d^* \leq p^*$$

$$d^* = p^* \quad \checkmark$$

dual

$$d^* = \max_{\alpha, \beta} \min_w L(w, \alpha, \beta)$$

- ① f and g should be convex and h should be affine
- ② g_i are (strictly) feasible $g_i(c) \leq 0 \quad \forall i$
- there must exist w^*, α^*, β^* and $d^* = p^* = L(w^*, \alpha^*, \beta^*)$
and w^*, α^*, β^* should satisfy KKT

$$\frac{\partial L}{\partial w} = 0 \quad \frac{\partial L}{\partial \alpha} = 0 \quad \frac{\partial L}{\partial \beta} = 0 \quad \boxed{\alpha_i^* g_i(c) \leq 0}$$

$g_i(c) \leq 0$

$\alpha^* \geq 0$

1960's Vladimir Vapnik & Alexey Chervonenkis

1961's $(x, y) \rightarrow$ kernel $\rightarrow$ transform a non-linear data to a higher dimension with linear separability

$$K(x, y) = \phi(x) \cdot \phi(y)$$

1968 VC dimension: Complexity of the model w.r.t the data point $\Rightarrow$ how can a model be generalized to unknown data.

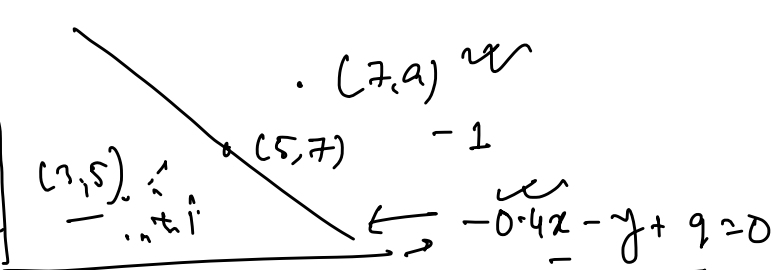
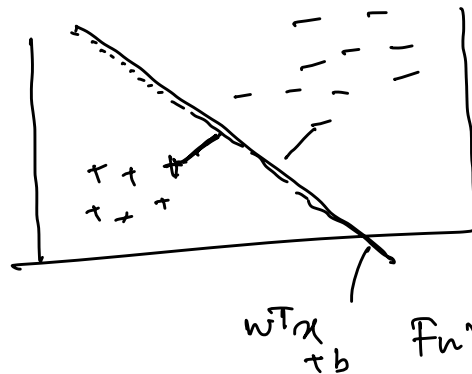
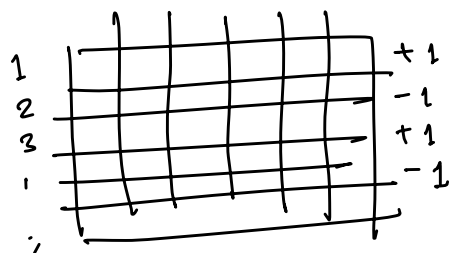
1990 AT&T bell lab

1992 COLT $\rightarrow$ kernel-trees

1995 $\rightarrow$ SVM $\rightarrow$ Soft-margin

98 $\rightarrow$ SMD $\rightarrow$ Scalable

08/09/22



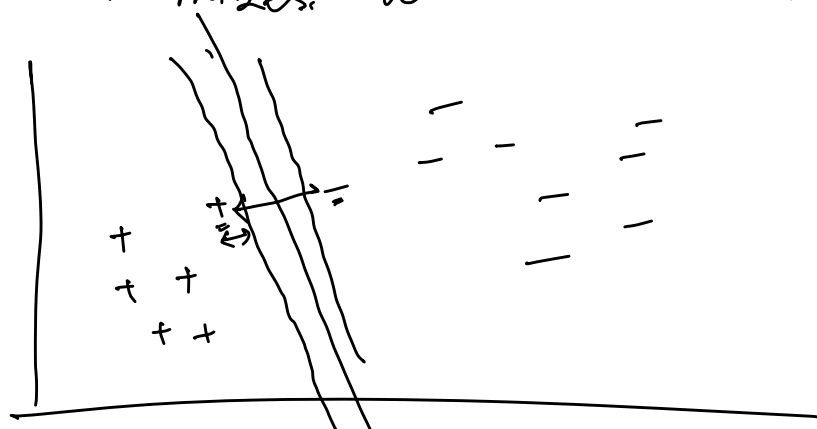
Functional Margin:

$$\hat{y}_i = y_i (w^T x_i + b)$$

Fun Margin

$$\begin{aligned} (3,5) &\rightarrow (2.8) \rightarrow 2.8 \times 2 \\ (5,7) &\rightarrow 0 \\ (7,9) &\rightarrow -2.8 \end{aligned}$$

Objective : TO Identify a line that
* maximizes the minimum functional margin.



$$\begin{aligned} -0.4x_i - y_i + 9 &= 0 \\ -0.8x_i - 2y_i + 18 &= 0 \end{aligned}$$

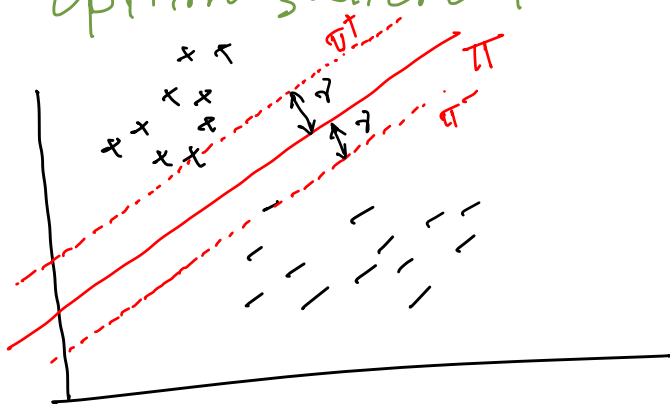
Functional margin is
"Scale-variant"

Geometric Margin

$$\hat{\gamma} = y_i \left(\frac{w^T x_i + b}{\|w\|} \right) \leftarrow \text{Scale invariant}$$

New objective : TO identify a line that maximizes the minimum geometric margin.

Optimization problem for Hard margin of SVM



$$\max_{w,b} \hat{\gamma} \quad \text{s.t.c.} \quad \text{geometric Margin}_i \geq \hat{\gamma} \quad i=1 \dots m$$

$$\begin{aligned} \max_{w,b} \quad & \hat{\gamma} / \|w\| \\ \text{s.t.c.} \quad & y_i (w^T x_i + b) \geq \hat{\gamma} \quad i=1 \dots m \end{aligned}$$

$$\begin{aligned} \max \quad & 1 / \|w\| \\ \text{s.t.c.} \quad & y_i (w^T x_i + b) \geq 1 \end{aligned}$$

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{s.t.} \quad y_i (w^T x_i + b) \geq 1$$

$$\Rightarrow -y_i (w^T x_i + b) + 1 \leq 0 \iff g_i(w)$$

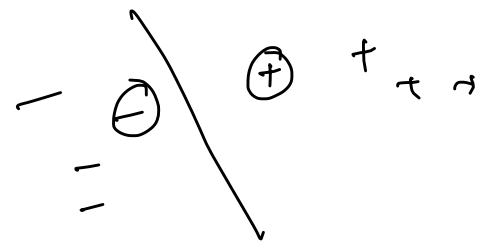
Instead of solving primal, we will solve dual, and w^*, α^*, b^* satisfying KKT

$$\alpha_i g_i(w) = 0$$

$$(i) \quad g_i(w) > 0 \Rightarrow \text{if } y_i (w^T x_i + b) < 1$$

$$\alpha_i > 0 \rightarrow \text{SV}$$

$$(ii) \quad \frac{g_i(w) \neq 0}{y_i (w^T x_i + b) \geq 1} \quad \alpha_i = 0 \quad g_i(w) \Rightarrow -ve$$



Support vectors = are the data points which are nearest to the decision boundary.

For SV, $\alpha_i > 0$

for other points $\alpha_i = 0$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^m \alpha_i (y_i (w^T x_i + b) - 1) \quad \text{--- (1)}$$

$$\mathcal{D}_d(\alpha) = \min_{w, b} \mathcal{L}(\quad)$$

$$\frac{\partial \mathcal{L}}{\partial w} \Rightarrow w - \sum_{i=1}^m \alpha_i y_i x_i = 0 \quad \text{--- (ii)}$$

$$\frac{\partial \mathcal{L}}{\partial b} \Rightarrow \sum_{i=1}^m \alpha_i y_i = 0 \quad \text{--- (iii)}$$

Substitute w obtained in (ii) to (1)

$$\frac{1}{2} \sum_{i=1}^m \alpha_i y_i x_i \sum_{j=1}^m \alpha_j y_j x_j - \left(\sum_{i=1}^m \alpha_i \left(y_i \left(\sum_{j=1}^m (\alpha_j y_j x_j) x_i + b \right) - 1 \right) \right)$$

$$= \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j (x_i \cdot x_j) - \left(\sum_{i=1}^m \alpha_i y_i \left(\sum_{j=1}^m (\alpha_j y_j x_j) x_i + b \right) \right) + \sum_{i=1}^m \alpha_i$$

$$= \sum_{i=1}^m \alpha_i + \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j (x_i \cdot x_j) - \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j (x_i \cdot x_j) - b \sum_{i=1}^m \alpha_i y_i$$

$$\leftarrow K_L(x_i, x_j)$$

Dual problem

$$\max_{\alpha} \min_{w, b} \mathcal{L}(\quad)$$

$$\max_{\alpha} \mathcal{D}_d(\quad)$$

$$= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (x_i \cdot x_j) - Q_D$$

$$\boxed{\begin{array}{ll} \max_{\alpha} & Q_D \\ \text{s.t.} & \alpha_i \geq 0 \\ & \sum_{i=1}^m \alpha_i y_i = 0 \end{array}}$$

α^*

if α^* is using in (1) $\rightarrow$ we will get w^*

$K(x_i, x_j)$
kernel

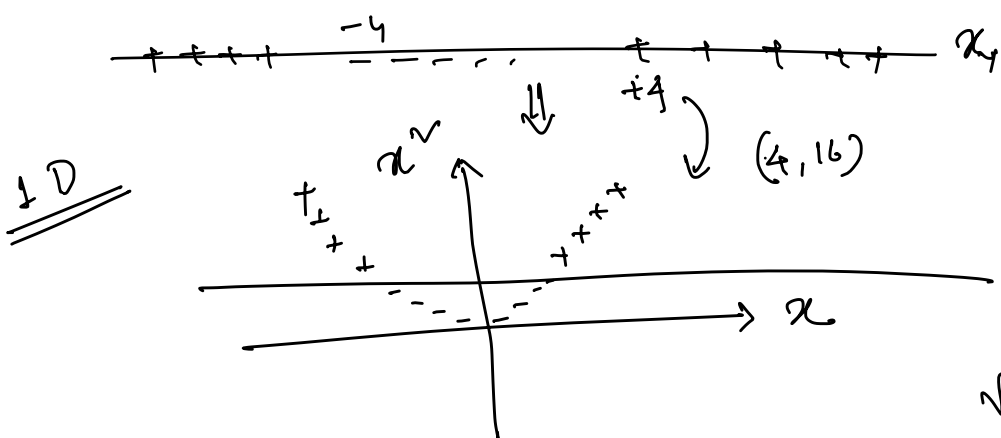
From $\alpha_i y_i (w) \geq 0 \rightarrow$ we will get w^*

Kernel: is a function that takes input as a pair of vectors in the original space and returns dot product of vectors in the transformed space.

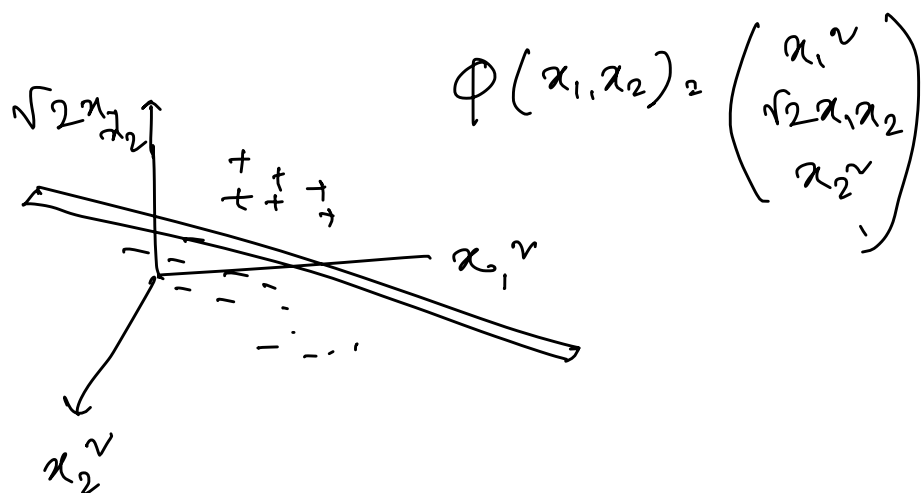
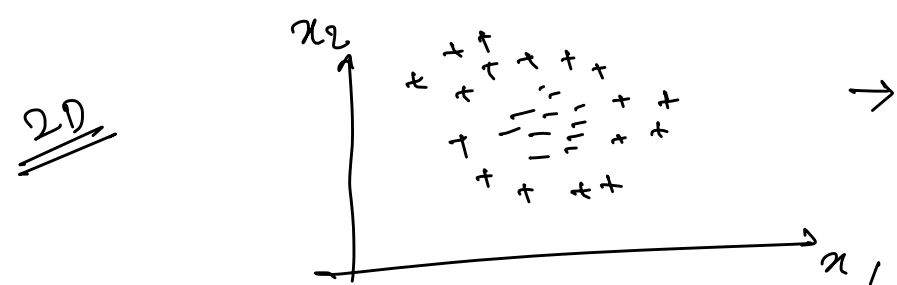
$$\begin{array}{c} \vec{a} \quad \vec{b} \\ (a_1, a_2) \quad (b_1, b_2) \end{array} \leftarrow 2$$

$$K(\vec{a}, \vec{b}) \rightarrow \underline{\underline{6}}$$

(4)



$$\Phi(x) = \begin{pmatrix} x \\ x^v \end{pmatrix}$$



$$a: (a_1, a_2)$$

$$b: (b_1, b_2)$$

$$\Phi(a) = \begin{pmatrix} a_1^v \\ \sqrt{2}a_1a_2 \\ a_2^v \end{pmatrix}$$

$$\Phi(b) = \begin{pmatrix} b_1^v \\ \sqrt{2}b_1b_2 \\ b_2^v \end{pmatrix}$$

$$\begin{aligned} \Phi(a) \cdot \Phi(b) &= \begin{pmatrix} a_1^v \\ \sqrt{2}a_1a_2 \\ a_2^v \end{pmatrix} \cdot \begin{pmatrix} b_1^v \\ \sqrt{2}b_1b_2 \\ b_2^v \end{pmatrix} \\ &= a_1^v b_1^v + 2a_1a_2b_1b_2 + a_2^v b_2^v \\ &= (a_1b_1 + a_2b_2)^v \\ &= \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \right)^v = \underline{\underline{K(a, b)}} \end{aligned}$$

